

PAPER_674: UQFF Compared to General LIGO Dataset: Strain, Phase, and Frequency Sweep

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Framework: Unified Quantum Field Framework (UQFF) v5.30

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Class: UQFFComparedToLIGOData | CP4 #258

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Domain: Gravitational Waves / LIGO

Abstract

We present the Unified Quantum Field Framework (UQFF) prediction for gravitational-wave strain $h_{\text{UQFF}}(f)$ across the LIGO sensitivity band (10–2000 Hz). The UQFF modifies the standard GR inspiral waveform through three multiplicative suppression factors: time-reversal zone f_{TRZ} , Schwarzschild-condensate S_{SCm} , and magnetic string modulation S_{U_m} . We demonstrate a systematic frequency-dependent suppression of order $\sim (1 - f_{\text{TRZ}}) \approx 0.9$ compared to GR, with phase shift $\Delta\phi = \kappa f_{\text{TRZ}} t_{\text{coal}}$.

Primary UQFF Equation

$$h_{\text{UQFF}}(f) = h_{\text{GR}}(f) \cdot (1 - f_{\text{TRZ}}) \cdot e^{-\rho_{\text{SCm}} r_s / k_B T_H} \cdot e^{-U_m 2\pi f / c^2}$$

Parameters

Parameter	Value	Description
m_1	$1.36 M_{\odot}$	Primary mass
m_2	$1.17 M_{\odot}$	Secondary mass
d	40 Mpc	Luminosity distance
f	10–2000 Hz	LIGO band

UQFF Constants

Constant	Value	Description
ρ_{UA}	$7.09 \times 10^{-36} \text{ kg/m}^3$	Universal Aether density

Constant	Value	Description
ρ_{SCm}	$7.09 \times 10^{-37} \text{ kg/m}^3$	Schwarzschild condensate
f_{TRZ}	0.1	Time-reversal zone factor
κ	$5 \times 10^{-4} \text{ day}^{-1}$	UQFF calibration constant
[SSq]	0.57	Superstring quenching factor
μ_J	$3.38 \times 10^{23} \text{ J}\cdot\text{m}$	Magnetic string coupling
γ	$5 \times 10^{-5} / 86400 \text{ s}^{-1}$	Aether oscillation decay

C++ Implementation

```
#include "UQFFComparedToLIGOData.h"
```

```
UQFFComparedToLIGOData obj;
auto result = obj.compute_primary(...); // primary equation
obj.simulate_over_M(M_start, M_end, dM, "output.csv");
```

Python CP4 Calculator

```
from CondensedPhysics2 import UQFFComparedToLIGODataCalculator

calc = UQFFComparedToLIGODataCalculator()
result = calc.compute() # returns all UQFF quantities
print(result)
```

References

LIGO Scientific Collaboration 2016 (GW150914); Abbott et al. 2019 (O3); UQFF: Session 173

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